Topic 7 – Respiration, Muscles and the Internal Environment

Candidates will be assessed on their ability to:

7.1	(i) understand the overall reaction of aerobic respiration as splitting of the respiratory substrate to release carbon dioxide as a waste product and reuniting hydrogen with atmospheric oxygen with the release of large amounts of energy (ii) understand that respiration is a stepped process, with each step controlled and catalysed by a specific intracellular enzyme *Names of specific enzymes are not required.*
7.2	understand the roles of glycolysis in aerobic and anaerobic respiration, including the phosphorylation of hexoses, the production of ATP by substrate level phosphorylation, reduced coenzyme, pyruvate and lactate *Details of intermediate stages and compounds are not required.*
7.3	understand the role of the link reaction and the Krebs cycle in the complete oxidation of glucose and formation of carbon dioxide (CO_2) by decarboxylation, ATP by substrate level phosphorylation, reduced NAD and reduced FAD by dehydrogenation (names of other compounds are not required) and that these steps take place in mitochondria, unlike glycolysis which occurs in the cytoplasm
7.4	understand how ATP is synthesised by oxidative phosphorylation associated with the electron transport chain in mitochondria, including the role of chemiosmosis and ATP synthase
7.5	understand what happens to lactate after a period of anaerobic respiration in animals
7.6	understand what is meant by the term *respiratory quotient* (*RQ*)
7.7	**CORE PRACTICAL 15** **Use an artificial hydrogen carrier (redox indicator) to investigate respiration in yeast.**
7.8	**CORE PRACTICAL 16** **Use a simple respirometer to determine the rate of respiration and RQ of a suitable material (such as germinating seeds or small invertebrates).**
7.9	know the way in which muscles, tendons, the skeleton and ligaments interact to enable movement, including antagonistic muscle pairs, extensors and flexors
7.10	(i) know the structure of a mammalian skeletal muscle fibre (ii) understand the structural and physiological differences between fast and slow twitch muscle fibres
7.11	understand the process of contraction of skeletal muscle in terms of the sliding filament theory, including the role of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase
7.12	(i) know the myogenic nature of cardiac muscle (ii) understand how the normal electrical activity of the heart coordinates the heartbeat, including the roles of the sinoatrial node (SAN), the atrioventricular node (AVN), the bundle of His and the Purkyne fibres (iii) understand how the use of electrocardiograms (ECGs) can aid in the diagnosis of abnormal heart rhythms

7.13	(i) be able to calculate cardiac output (ii) understand how variations in ventilation and cardiac output enable rapid delivery of oxygen to tissues and the removal of carbon dioxide from them, including how the heart rate and ventilation rate are controlled and the roles of the cardiovascular control centre and the ventilation centre in the medulla oblongata
7.14	understand the role of adrenaline in the fight or flight response
7.15	**CORE PRACTICAL 17** **Investigate the effects of exercise on tidal volume, breathing rate, respiratory minute ventilation, and oxygen consumption using data from spirometer traces.**
7.16	(i) understand what is meant by the terms *negative feedback* and *positive feedback control* (ii) understand the principle of negative feedback in maintaining systems within narrow limits
7.17	understand what is meant by the term *homeostasis* and its importance in maintaining the body in a state of dynamic equilibrium during exercise, including the role of the hypothalamus in thermoregulation
7.18	know the gross and microscopic structure of the mammalian kidney
7.19	understand how urea is produced in the liver from excess amino acids (*details of the ornithine cycle are not required*) and how it is removed from the bloodstream by ultrafiltration
7.20	understand how solutes are selectively reabsorbed in the proximal tubule and how the loop of Henle acts as a countercurrent multiplier to increase the reabsorption of water
7.21	understand how the pituitary gland and osmoreceptors in the hypothalamus, combined with the action of antidiuretic hormone (ADH), bring about negative feedback control of mammalian plasma concentration and blood volume
7.22	understand how genes can be switched on and off by DNA transcription factors, including the role of peptide hormones acting extracellularly and steroid hormones acting intracellularly